

Quantized Retrieval-Augmented Medical Vision-Language Diagnosis

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Abstract—This paper proposes a multimodal medical diagnosis pipeline that aligns medical images and clinical reports in a shared latent space, compresses the latent representation for efficient retrieval, and augments diagnostic reasoning with similar medical cases.

Index Terms—medical vision-language model, retrieval-augmented generation, quantization, FAISS, explainable diagnosis

I. INTRODUCTION

Medical diagnosis from imaging data benefits from models that jointly reason over visual findings, historical reports, and related cases. We study a quantized retrieval-augmented vision-language pipeline for MRI, CT, and histopathology diagnosis.

II. RELATED WORK

A. Medical Vision-Language Learning

Review biomedical image-text representation learning, contrastive alignment, and clinical report modeling.

B. Retrieval-Augmented Medical Diagnosis

Discuss similar-case retrieval, case-based reasoning, and retrieval-augmented generation for clinical decision support.

C. Quantized Vector Retrieval

Discuss product quantization, vector quantization, and compressed nearest-neighbor search.

III. METHOD

A. Image and Text Encoders

The image encoder maps MRI, CT, or histopathology inputs into visual embeddings. The text encoder maps clinical reports into biomedical text embeddings.

B. Cross-Modal Alignment

Image and report embeddings are projected into a shared latent space and optimized with a symmetric InfoNCE objective:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(z_i, z_t)/\tau)}{\sum_j \exp(\text{sim}(z_i, z_j)/\tau)}. \quad (1)$$

C. Quantized Retrieval

The aligned medical embeddings are compressed before indexing. At inference time, a query image embedding retrieves similar cases from a vector database.

D. Diagnosis and Report Generation

Retrieved cases provide context for disease classification and report generation.

IV. EXPERIMENTS

A. Datasets

Candidate datasets include BraTS, FastMRI, IXI, MIMIC-CXR, CheXpert, CAMELYON, and PANDA.

B. Metrics

Evaluate classification accuracy, retrieval recall, report quality, calibration, latency, and compression ratio.

V. DISCUSSION

Discuss explainability, uncertainty estimation, privacy, edge deployment, and clinical limitations.

VI. CONCLUSION

We outline a research direction for efficient and explainable medical vision-language diagnosis through quantized retrieval-augmented reasoning.

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